Breast Cancer Survival Analysis

Prognostic Factors and Patient Risk Stratification

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Cohort: TCGA Breast Invasive Carcinoma (BRCA)

Graphical Abstract

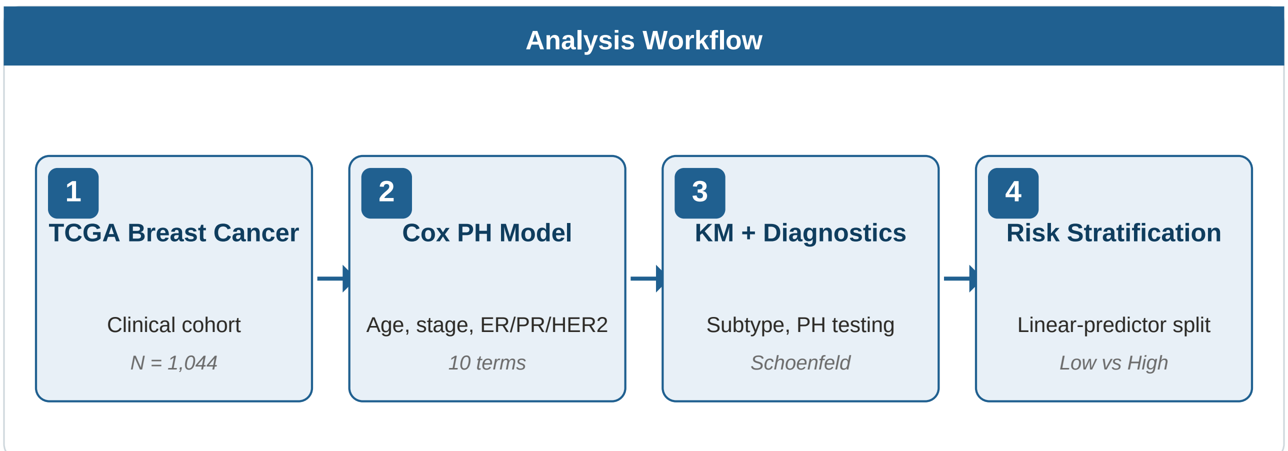

Figure 1. *Workflow summary: TCGA BRCA clinical cohort (N = 1,044) → multivariable Cox proportional hazards model on standard clinical covariates → survival diagnostics and proportional-hazards testing → Cox-derived risk score with median split into Low- and High-risk groups.*

Executive Summary

• **Cohort.** 1044 patients with breast invasive carcinoma; 104 deaths during follow-up (10.0% event rate; 90% censored).
• **Significant prognostic factors (BH-FDR < 0.05).** Age at diagnosis (HR 1.04 per year), AJCC Stage III (HR 3.00), and Stage IV (HR 6.14) all associated with higher hazard. Receptor-status terms (ER/PR/HER2) did not survive FDR after adjusting for age and stage.
• **Subtype effect.** Significant differences across the four molecular subtypes (log-rank $p = 3.3 \times 10^{-4}$); HER2+ and Triple-Negative subtypes show the lowest 5-year landmark OS (~70%).
• **Risk stratification.** Cox-derived median split separates patients clearly: High-risk vs Low-risk hazard ratio = 3.71 (95% CI 2.11–6.53), log-rank $p = 1.1 \times 10^{-6}$. 5-year OS: 70.6% (High) vs 92.8% (Low).
• **Caveats.** Heavy censoring (90%) and short median follow-up (1.33 yr) mean late-tail KM estimates are unreliable — landmark rates are the primary survival readout. Events-per-variable = 5.9 (< 10) flags potential overfitting of the C-index (0.81). PH assumption violated for stage (global $p = 3.9 \times 10^{-4}$); HRs represent time-averaged effects. **Discovery only — no external validation cohort.**

Methods

Data source. Clinical data for The Cancer Genome Atlas Breast Invasive Carcinoma (TCGA-BRCA) cohort were obtained programmatically via the `RTCGA.clinical` Bioconductor package (release 20151101). The endpoint analysed was overall survival (OS): time from diagnosis to death from any cause, with patients alive at last follow-up censored.

Cohort filtering and derived variables. Patients with missing survival time or event status, or with time = 0, were excluded (final analytic cohort N = 1044). Time was converted from days to years (days ÷ 365.25). AJCC pathologic stage was simplified to Stage I–IV. Molecular subtype was derived from receptor status: HR+/HER2– (ER or PR Positive, HER2 Negative), HR+/HER2+ (ER or PR Positive, HER2 Positive), HER2+ (ER & PR Negative, HER2 Positive), Triple Negative (ER, PR, HER2 all Negative).

Statistical methods. Kaplan–Meier survival curves with 95% confidence intervals were computed for the overall cohort and stratified by molecular subtype; group comparisons used the log-rank test. A multivariable Cox proportional hazards model was fit on the standard clinical panel (age, AJCC stage, ER/PR/HER2 status, molecular subtype). The skill's collinearity check identified molecular subtype as collinear with ER status (Cramér's V = 0.95) and dropped it from the fitted model. **Final fitted model**: age, stage, ER status, PR status, HER2 status (10 coefficient terms total).

Reference levels. stage = Stage II (N = 582); ER = Positive (N = 773); PR = Positive (N = 668); HER2 = Negative (N = 540). The most common level was used as reference for each categorical covariate.

Proportional hazards diagnostics. The PH assumption was assessed using Schoenfeld residual tests per covariate and globally. The concordance index (C-index) was computed for overall model discrimination. Multiple testing correction across the 10 coefficient p-values used the Benjamini–Hochberg method (FDR < 0.05).

Risk stratification. A patient-level risk score was computed as the Cox linear predictor $\beta^T x$, where β are the fitted coefficients and x is the covariate vector. Patients were split at the median into Low- and High-risk groups. Group survival was compared with the log-rank test and a univariate Cox model.

Software. All analyses were performed in R 4.4.2 using `survival` v3.5+, `survminer` v0.4.9+, and `RTCGA.clinical`. Figures were generated via the `survminer` `ggsurvplot` family.

Results

Cohort Characteristics

Characteristic	Value
Total patients	1044
Deaths (events)	104 (10.0%)
Censored	940 (90.0%)
Age, mean (SD)	58.4 (13.2) yr
Age, range	26–90 yr
Median follow-up (reverse KM)	1.33 yr
Max observation time	19.35 yr
Stage I	179 (17.1%)
Stage II	582 (55.7%)
Stage III	244 (23.4%)
Stage IV	20 (1.9%)
Stage missing	19 (1.8%)
HR+/HER2-	579 (55.5%)
HR+/HER2+	112 (10.7%)
HER2+	34 (3.3%)
Triple Negative	115 (11.0%)
Subtype missing	204 (19.5%)

Table 1. *Cohort characteristics for the analytic TCGA BRCA dataset.*

Overall Survival

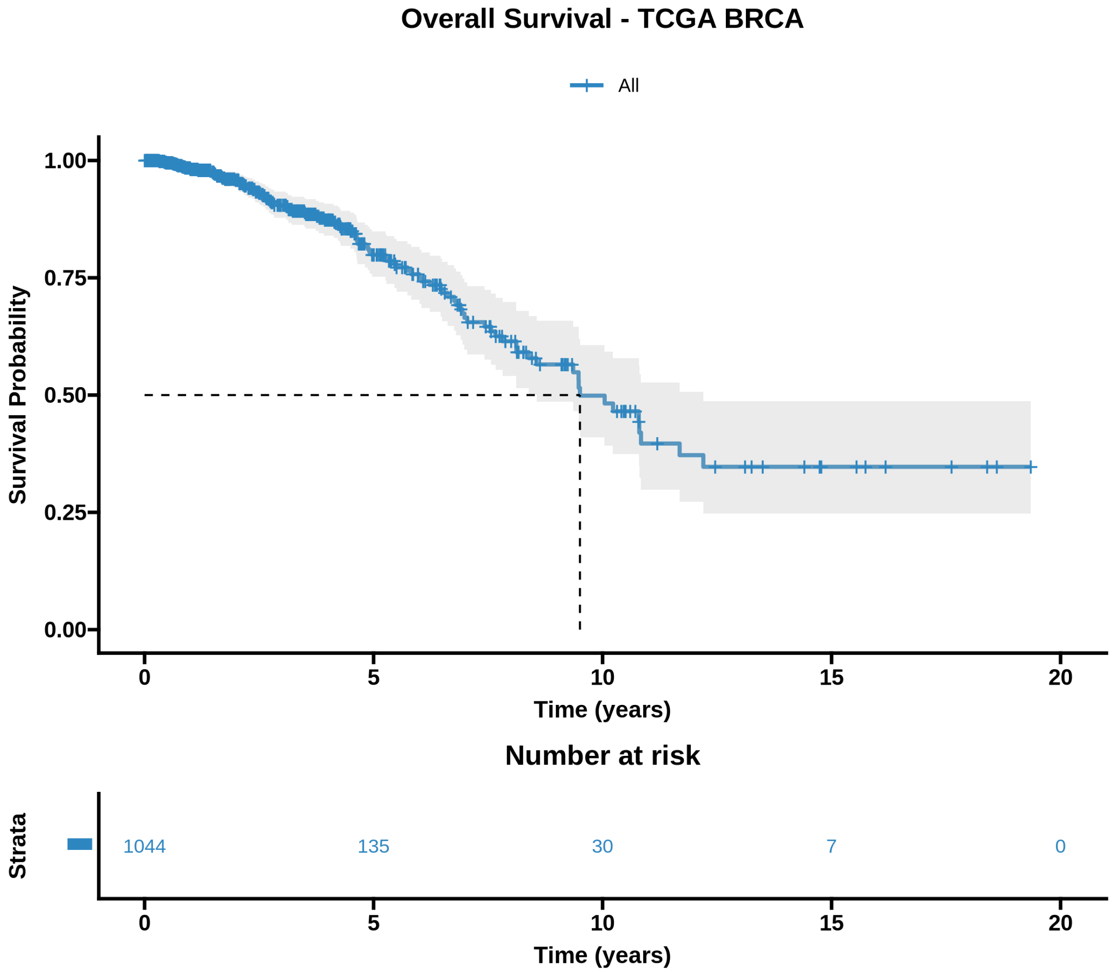

Figure 2. *Kaplan–Meier overall survival curve for the full cohort (N = 1044). Shaded band = 95% CI. The risk table shows numbers at risk over time. With 90% censoring, the late-tail estimates (where N at risk < 100) are unreliable; landmark rates below are the recommended summary.*

Timepoint	OS rate	95% CI	N at risk
1 year	98.1%	97.1%–99.2%	579
3 years	90.5%	87.8%–93.3%	304
5 years	79.9%	75.2%–84.9%	135

Table 2. *Landmark overall-survival rates at 1, 3, and 5 years, with 95% CIs.*

Survival by Molecular Subtype

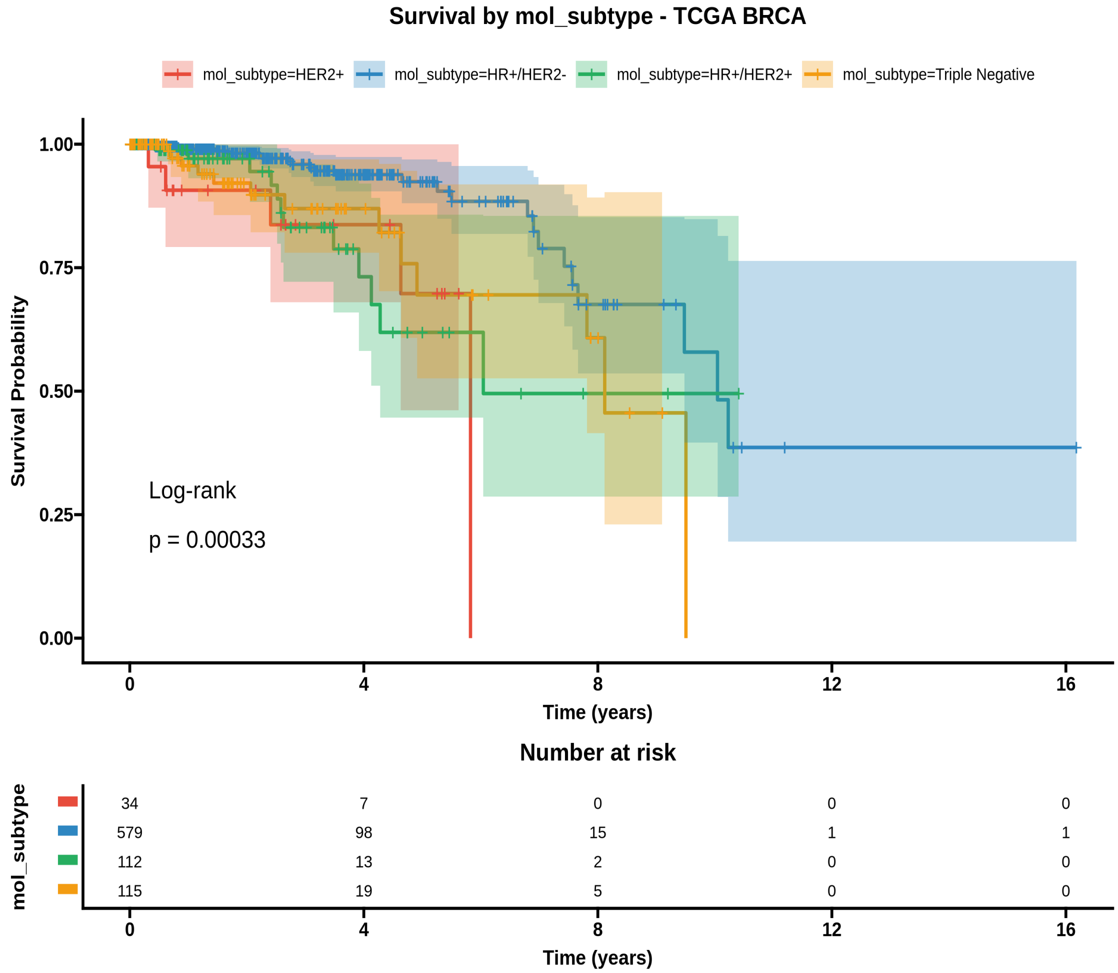

Figure 3. *Kaplan–Meier curves stratified by molecular subtype (N = 840 with subtype data). Log-rank p = 3.3 × 10^{-4}. HR+/HER2– (luminal-A-like, the most common subtype) shows the best survival; HER2+ and Triple-Negative subtypes diverge with worse outcomes.*

Subtype	N	1-yr OS	3-yr OS	5-yr OS
HR+/HER2-	579	99.0%	95.9%	92.4%
HR+/HER2+	112	97.1%	83.2%	61.9%
HER2+	34	90.7%	83.7%	69.8%
Triple Negative	115	95.6%	87.0%	69.5%

Table 3. *Landmark OS rates by molecular subtype. HER2+ has wide CIs due to small N (34).*

Multivariable Cox Model

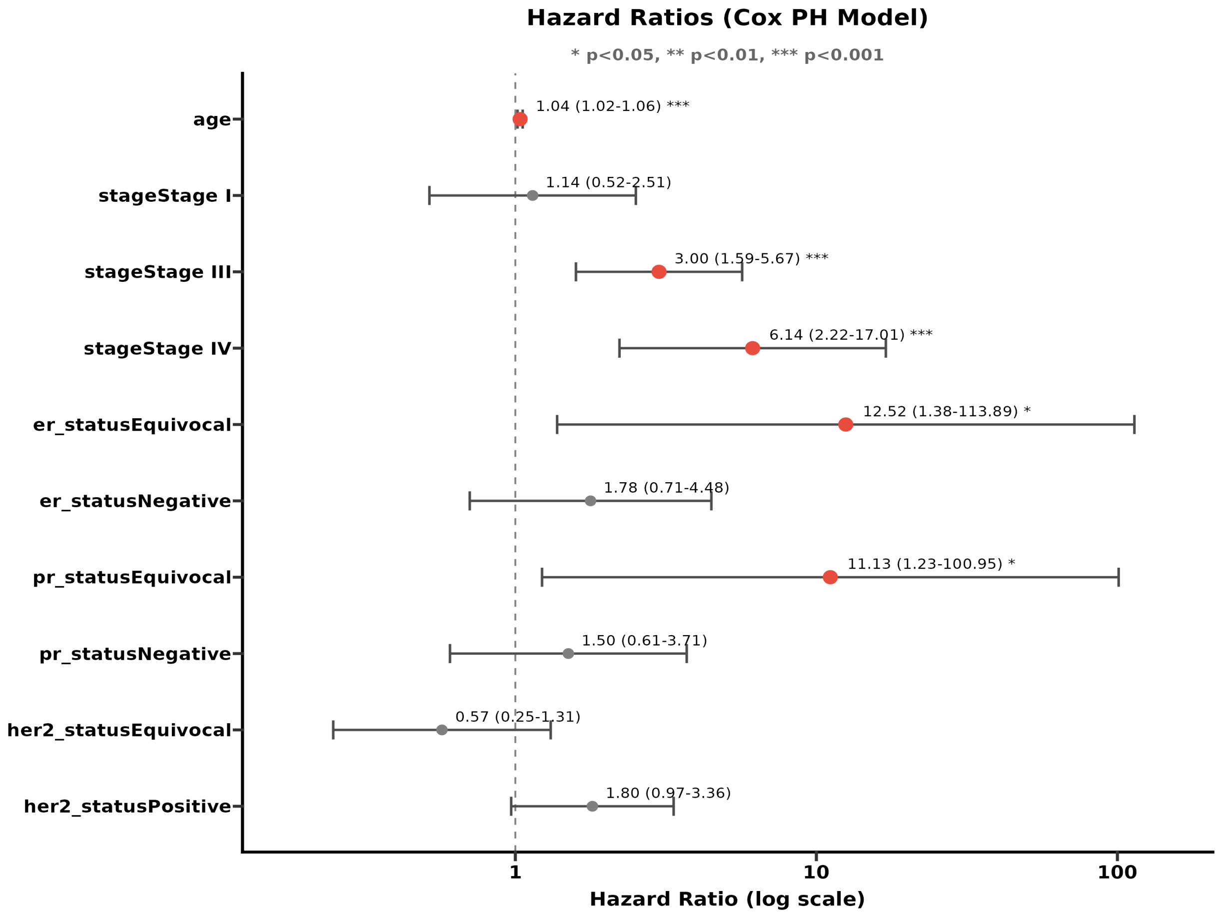

Figure 4. *Forest plot of hazard ratios from the multivariable Cox model. Points are HR estimates; whiskers are 95% CIs (log scale). The vertical reference line at HR = 1 marks no effect. Significant covariates after BH-FDR are age, Stage III, and Stage IV.*

WARNING. Global proportional hazards test p = 3.9×10^{-4} (Schoenfeld residuals). The PH assumption is violated for stage (p = 0.047). Reported hazard ratios should be interpreted as *time-averaged effects* and may be misleading.

Covariate	HR (95% CI)	Raw p	BH-adj p
Age (per year)	**1.04 (1.02–1.06)**	0.0003	**0.002**
Stage I (vs II)	1.14 (0.52–2.51)	0.743	0.743
Stage III (vs II)	**3.00 (1.59–5.67)**	0.0007	**0.002**
Stage IV (vs II)	**6.14 (2.22–17.01)**	0.0005	**0.002**
ER Equivocal (vs Positive)	12.52 (1.38–113.89)	0.025	0.062
ER Negative (vs Positive)	1.78 (0.71–4.48)	0.222	0.278
PR Equivocal (vs Positive)	11.13 (1.23–100.95)	0.032	0.064
PR Negative (vs Positive)	1.50 (0.61–3.71)	0.380	0.422
HER2 Equivocal (vs Negative)	0.57 (0.25–1.31)	0.186	0.266
HER2 Positive (vs Negative)	1.80 (0.97–3.36)	0.063	0.105

Table 4. *Cox model coefficients.* ***Bold*** *rows are significant at BH-FDR < 0.05. The ER- and PR-Equivocal HRs have extreme widths (1.4–113.9 and 1.2–101.0) due to tiny reference-group sizes (N = 2 and 4 respectively); these estimates are unstable.* ***PH assumption violated for stage (p = 0.047); HRs are time-averaged.*** *C-index = 0.809; events-per-variable = 5.9 (below the recommended ≥ 10 threshold — C-index may be optimistically biased).*

Proportional Hazards Diagnostics

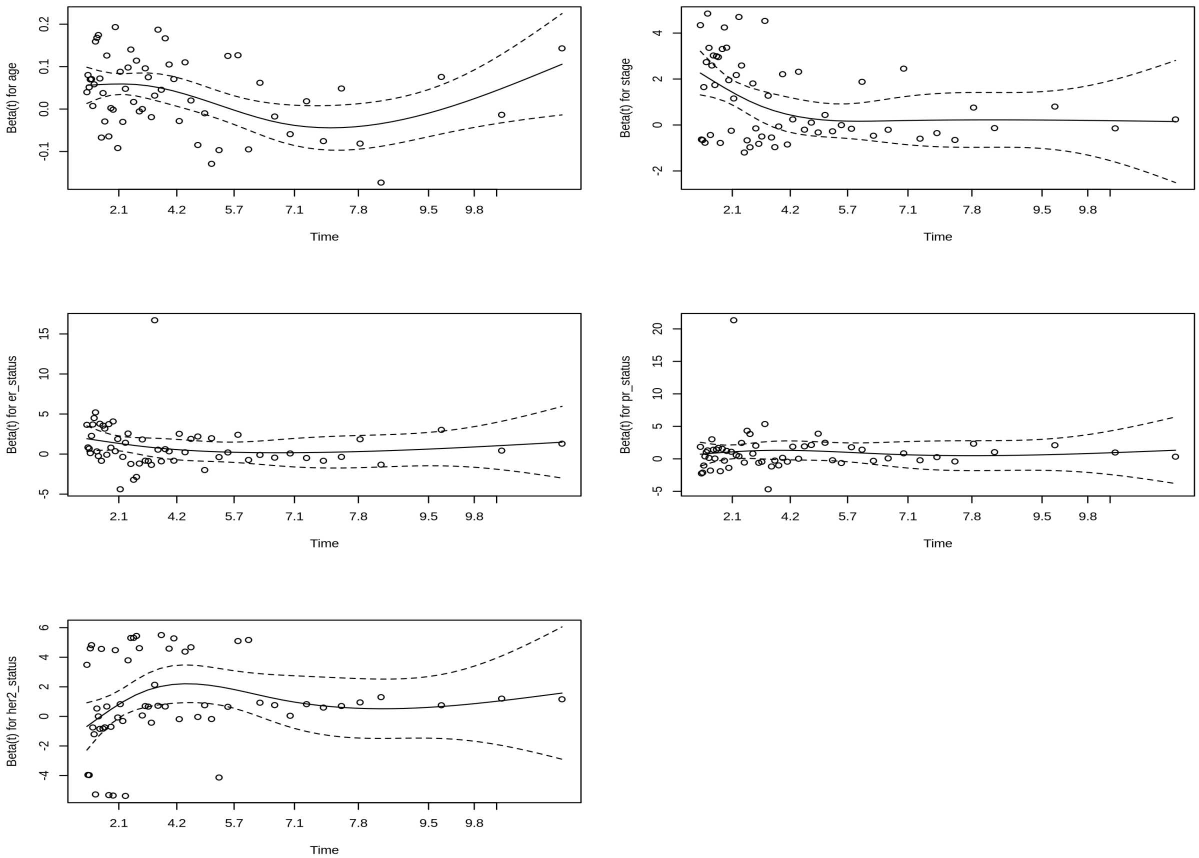

Figure 5. *Schoenfeld residual plots, one panel per covariate. Under a valid PH assumption the smoothed residual curve should be approximately flat at zero. The global PH test rejected the assumption (chi-sq = 32.05 on 10 df, p = 3.9 × 10^{-4}), driven primarily by* ***stage*** *(chi-sq = 7.95 on 3 df, p = 0.047). Possible remedies include stratifying by stage in the Cox model or fitting time-varying coefficients; both are out of scope for this discovery-only analysis.*

Covariate	chi-sq	df	p
Age	3.17	1	0.075
Stage	**7.95**	3	**0.047**
ER status	3.26	2	0.196
PR status	3.79	2	0.151
HER2 status	3.90	2	0.142
GLOBAL	**32.05**	10	**0.0004**

Table 5. *Schoenfeld residual test for the proportional hazards assumption. Bold rows reject PH (p < 0.05).*

Risk Stratification

Patients were stratified by the median of the Cox-derived linear predictor, yielding two groups of equal size: **Low-risk (N = 432)** and **High-risk (N = 432)**. 180 patients with missing covariates required for the Cox model could not be assigned a risk score and were excluded from this stratification.

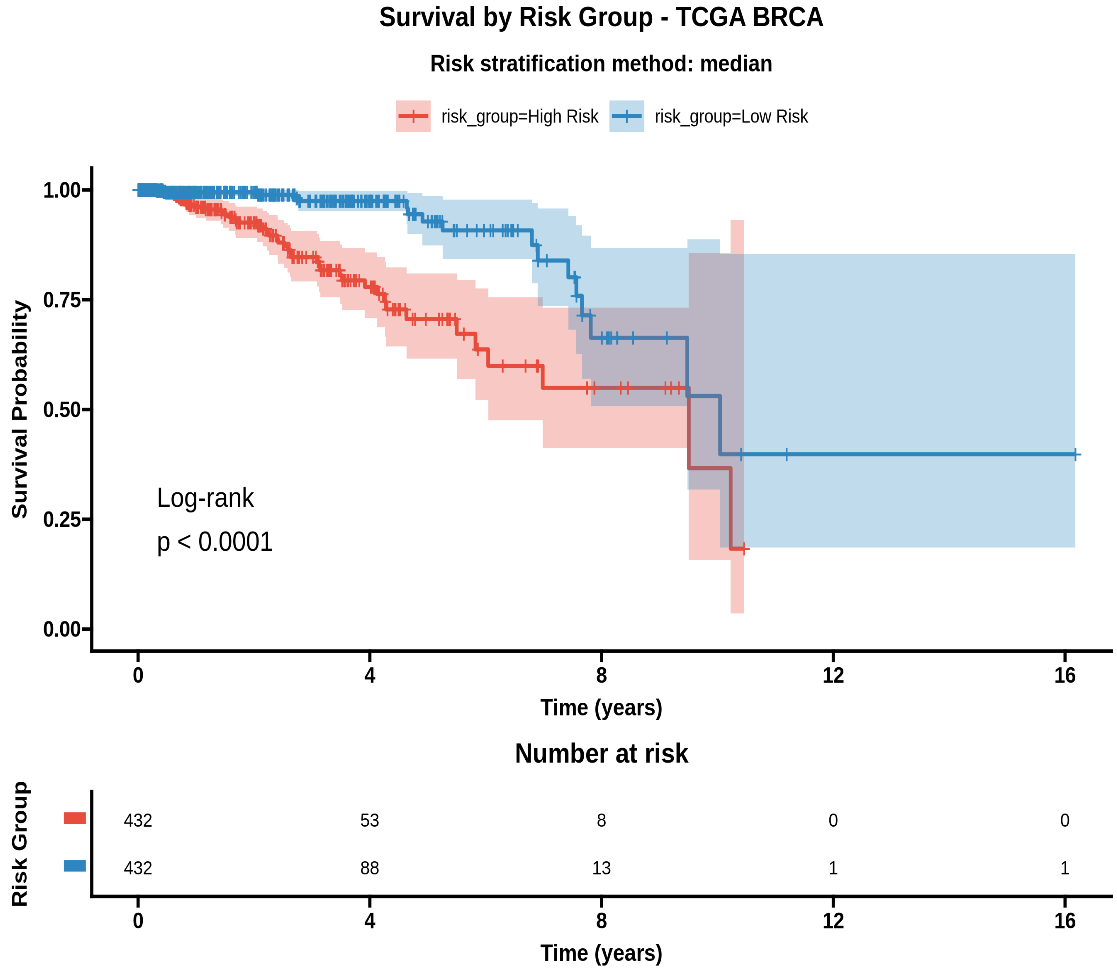

Figure 6. *Kaplan–Meier curves for Low- and High-risk groups defined by median split of the Cox linear predictor. Strong separation: log-rank p = 1.1 × 10^{-6}, hazard ratio (High vs Low) = 3.71 (95% CI 2.11–6.53).*

Risk Group	N	Events	1-yr OS	3-yr OS	5-yr OS
Low Risk	432	17	99.4%	97.5%	92.8%
High Risk	432	42	96.1%	84.7%	70.6%

Table 6. *Landmark survival by risk group. At 5 years the absolute difference in OS is 22.3 percentage points (92.8% vs 70.6%).*

Conclusions

- **Significant prognostic factors (BH-FDR < 0.05) in the TCGA-BRCA cohort are age and AJCC stage (III and IV).** Each additional year of age was associated with a 3.7% increase in hazard (HR 1.04, 95% CI 1.02–1.06). Stage III patients had ~3-fold higher hazard than Stage II (HR 3.00, 95% CI 1.59–5.67); Stage IV ~6-fold (HR 6.14, 95% CI 2.22–17.01) but with wide CIs due to small N (20 Stage-IV patients).
- **Molecular subtype effect is real but largely captured by receptor status.** The four subtypes differ significantly at the KM level (log-rank $p = 3.3 \times 10^{-4}$) and individually vs HR+/HER2– (univariate HRs 2.5–4.4). However, molecular subtype was algorithmically collinear with ER status (Cramér's V = 0.95) and dropped from the multivariable model. In a multivariable setting controlling for stage, receptor terms did not survive FDR — consistent with stage being the dominant axis of prognostic variation in this cohort.
- **Risk stratification works.** The Cox-derived median split separates patients with HR = 3.71 (95% CI 2.11–6.53) and an absolute 5-year OS difference of 22.3 percentage points. This score is internally derived; performance on independent cohorts is unknown.
- **Caveats and reliability.** The C-index of 0.81 is encouraging but with EPV = 5.9 (below the recommended ≥ 10) it is **potentially overfitted** to this dataset. The PH assumption is violated for stage (Schoenfeld p = 0.047; global $p = 3.9 \times 10^{-4}$); reported HRs are time-averaged. Heavy censoring (90%) and a 1.33-year median follow-up against a 19.3-year maximum suggest incomplete follow-up records for many censored patients; landmark rates were reported instead of median OS, which was “not reached” for all four molecular subtypes.
- **Selection bias risk.** Patients with missing molecular-subtype (N = 204) or HER2-status (N = 168) data had ~4× higher event rates than those with complete data (Fisher $p \sim 10^{-11}$). 180 patients were excluded from the Cox model and risk stratification, so the analysis is implicitly restricted to a healthier subset.
- **Discovery only.** This analysis used a single cohort (TCGA-BRCA) without external validation. The risk score and the prognostic effect sizes should be regarded as hypothesis-generating; validation on an independent breast-cancer cohort (e.g. METABRIC) is required before any clinical interpretation.

Output Files

All intermediate outputs are saved alongside this report. **Figures:** km_overall, km_stratified, km_risk_groups, forest_plot, schoenfeld_diagnostics (PNG + SVG). **Tables:** cox_coefficients.csv (with BH-adjusted p), landmark_by_risk.csv, landmark_by_subtype.csv, ph_assumption_test.csv, risk_group_summary.csv. **Data:** risk_scores.csv (patient-level linear predictor and group), clinical_annotated.csv (full clinical with risk groups), survival_model.rds (R analysis object for reproducibility).